

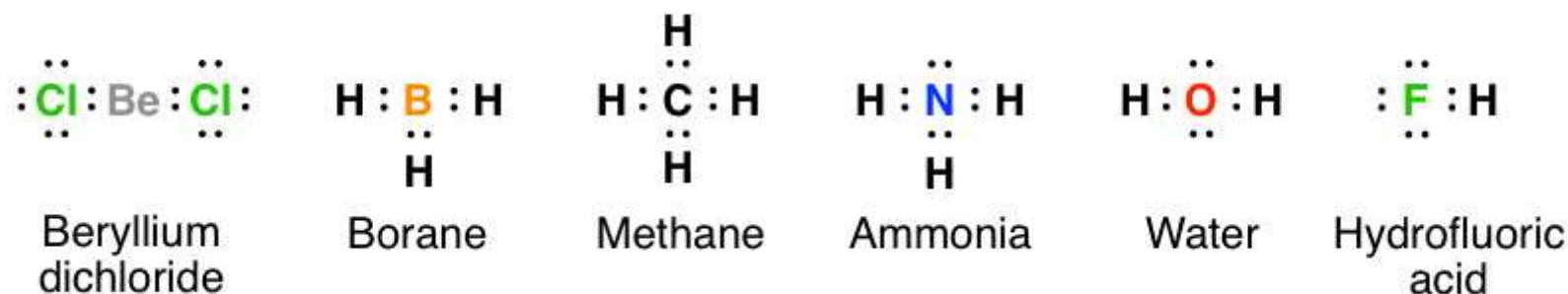
# Lewis Structures

There are essentially three uses for Lewis structures. They're good at helping you get acquainted with the placement of electrons around atoms, helping to visualize molecular geometry, and finally to remember the location of the lone pairs. As you'll see in Org 1, the geometry of molecules as well as the placement of lone pairs has a huge impact on reactivity, so using Lewis structures to get yourself reacquainted with these principles would be a worthwhile exercise.

## What are Lewis structures?

They are a way of drawing molecules that show the placement of electrons between atoms. For instance, here are the lewis dot structures of beryllium dichloride, borane, methane, ammonia, water, and hydrofluoric acid.

### The Full Lewis

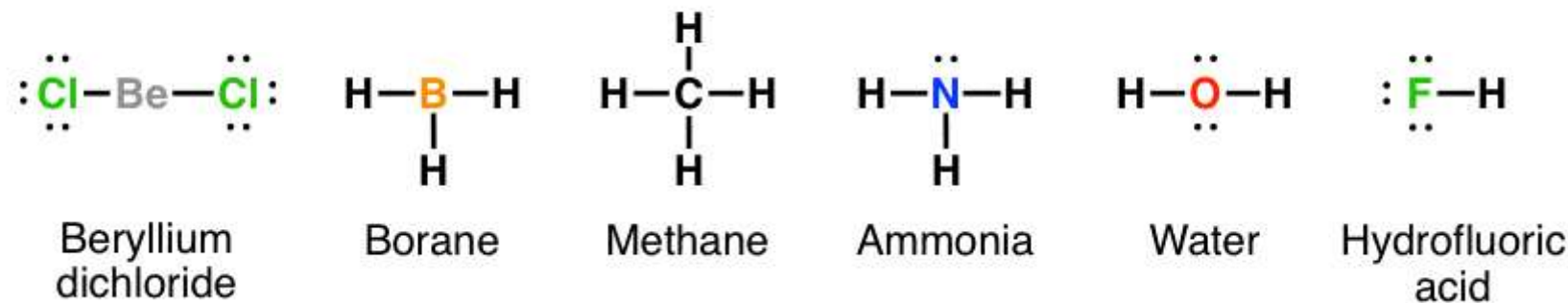


**The advantage of the full Lewis is that it allows you to see where all the electrons are and to determine if each atom obeys the octet rule.** You can see clearly that molecules can have both bonding electrons, which are shared between atoms and non-bonding electrons, otherwise known as lone pairs.

The full Lewis is kind of like training wheels on a bicycle. It's useful when you are just getting started and feeling uneasy about this business of atoms, electrons, and molecules, and want to determine for yourself that the octet rule is indeed a widespread phenomenon that (most) molecules abide by [the exceptions here being beryllium and boron, which are electron-deficient].

However, you will quickly realize that it is actually kind of a pain to draw a full Lewis structure. Once you become familiar with the basics of drawing molecules, you'll find it's much less work to simply draw a line where the bond is, as shown below. This also has the advantage that it is much easier to show geometry using the line bonds, because it's less cluttered.

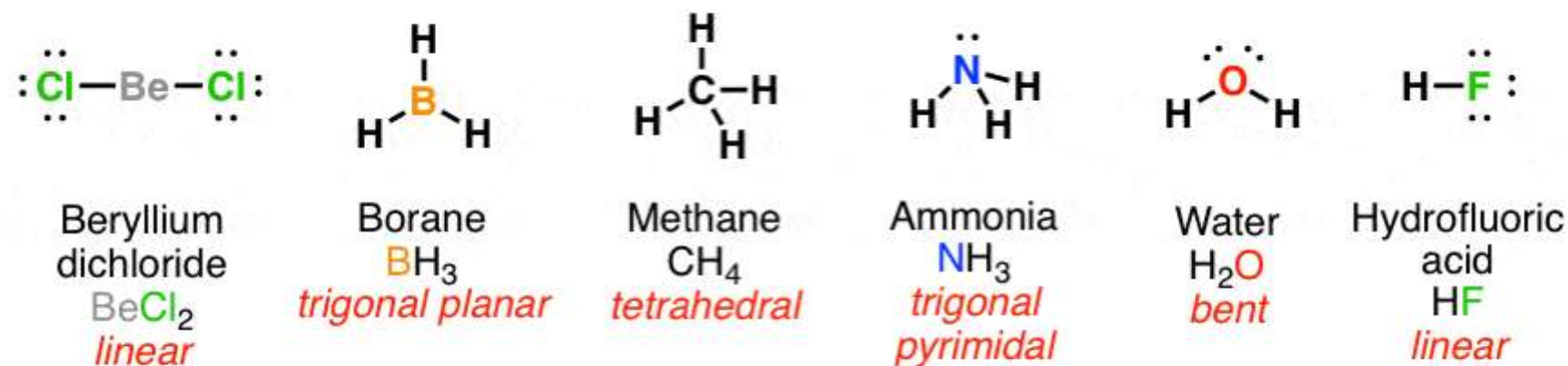
### The "Half-Lewis"



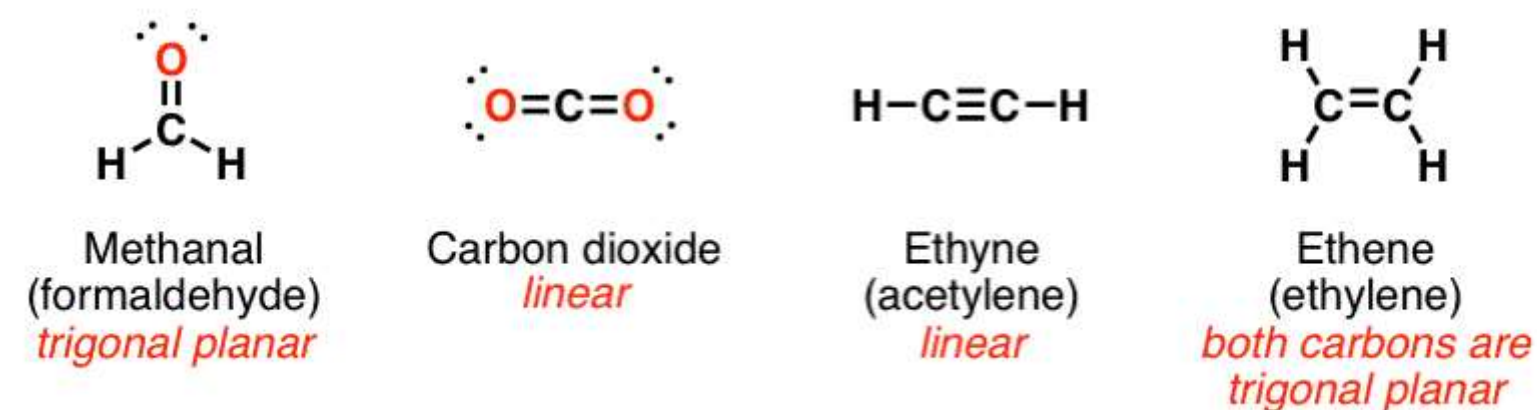
This brings us to our second point. **Electron pairs repel – this applies both to bonding electrons and electrons in lone pairs. So molecules will adopt a geometry which maximizes the distance between them.** This is why methane is tetrahedral, (internal angles 109°) and not square planar (internal angles 90°), and water is bent, not linear. You might recall this is referred to as VSEPR (valence shell electron pair repulsion).

Drawing out molecular geometry using a full Lewis makes for an extremely cluttered drawing. That's why we drop the full Lewis for the half-Lewis, move the lines around, and just leave the electron pairs in.

Line drawings can be used to depict molecular geometry:

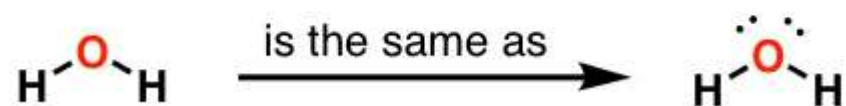
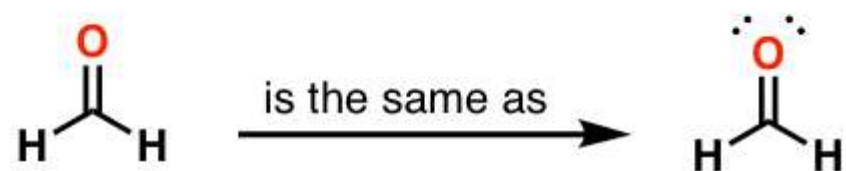


Also applies to molecules with multiple bonding:

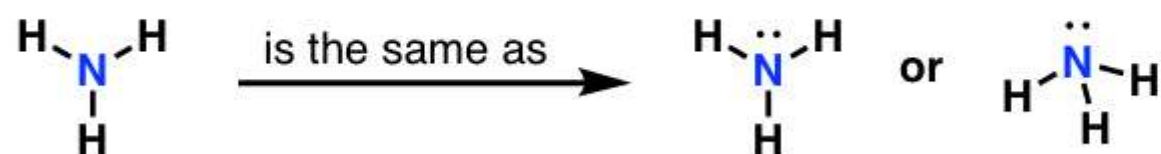


Now there is even a second tier of laziness. It's less work to just drop drawing the electron pairs altogether. This is by far the most common way molecules are drawn. Let's look at a few examples.

Further examples of laziness (or as we would prefer to put it, *shorthand*)



*The lone pairs haven't disappeared, it's just slightly less work if you omit drawing them in*



**You're supposed to know that the lone pairs are still there, even though they're not drawn in.** Think of them like chemical stick figures. For instance, [xkcd](#) tends not to draw faces, feet, or hands, but that doesn't mean the characters he draws are implied to be faceless amputees. It's just quicker to draw stick figures. **This is important because in organic chemistry, lone pairs often don't just sit around. As you'll see later, lone pairs are [nucleophiles](#) – they participate in a host of chemical reactions. So it's crucial to know that they're there, even if they're not drawn in.**